

Lecture 11: Solving Linear Programming Problems

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Lecturer: Mitchell Scott

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11.1 Geometry of the Objective Function

We know that the objective function of any linear programming problem could be written as

$$\mathbf{c}^\top \mathbf{x}.$$

This has to equal something, so we say that $\mathbf{c}^\top \mathbf{x} = k$. Then the goal is to optimize this hyperplane. We can think of this as finding the first set of feasible solutions that intersect the solution hyperplane.

Example 11.1 *Consider the linear programming problem*

$$\text{maximize } z = 4x + 3y \text{ subject to}$$

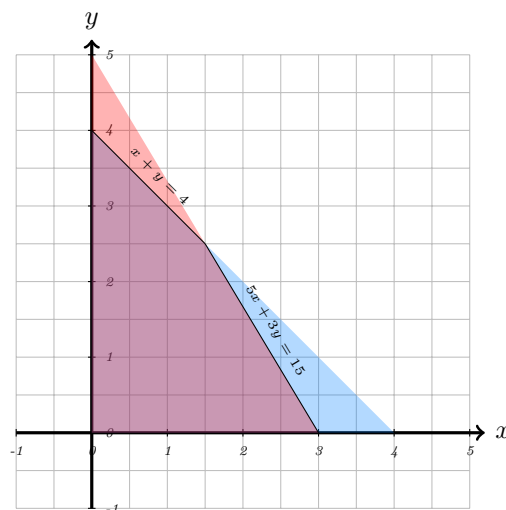
$$x + y < 4$$

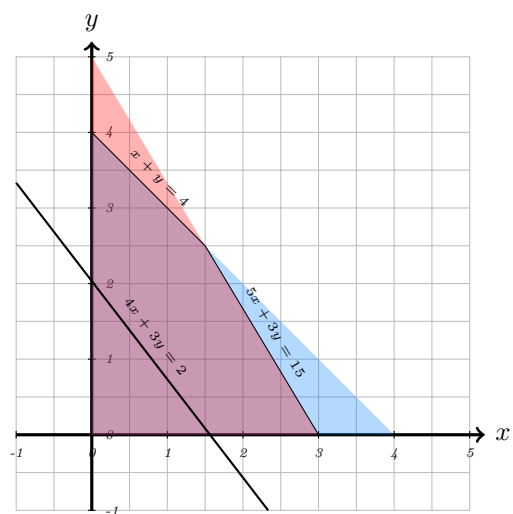
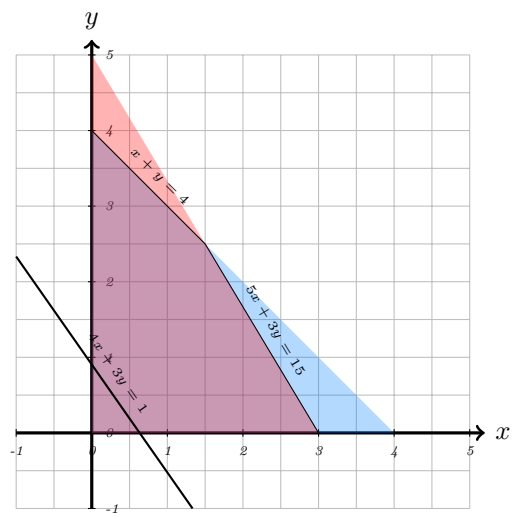
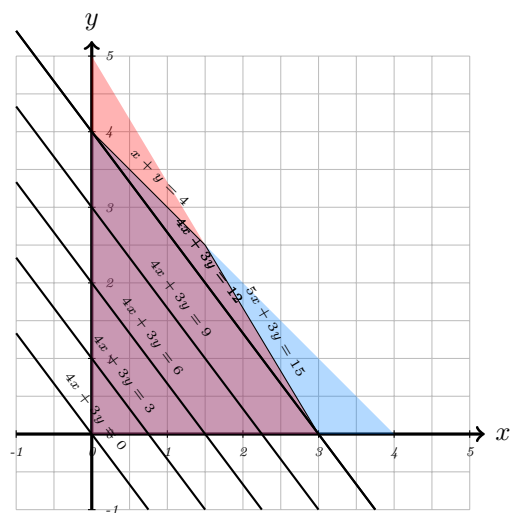
$$5x + 3y < 15$$

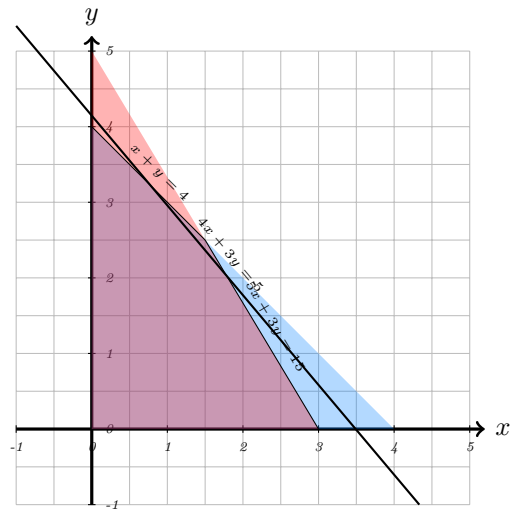
$$x > 0$$

$$y > 0$$

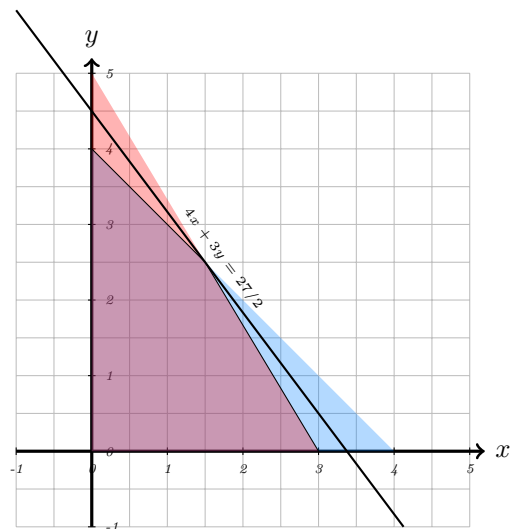
The set of feasible solutions (the shaded region) and the hyperplane





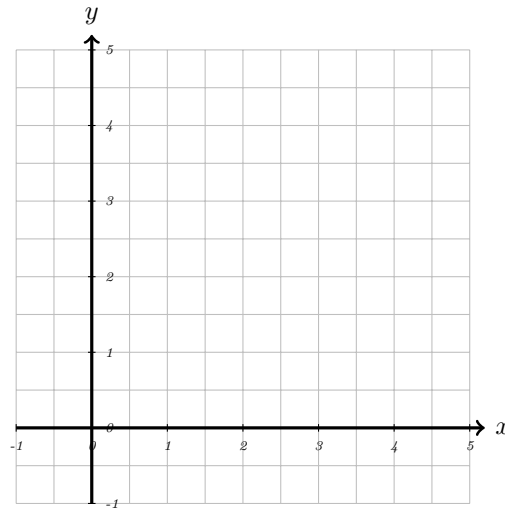
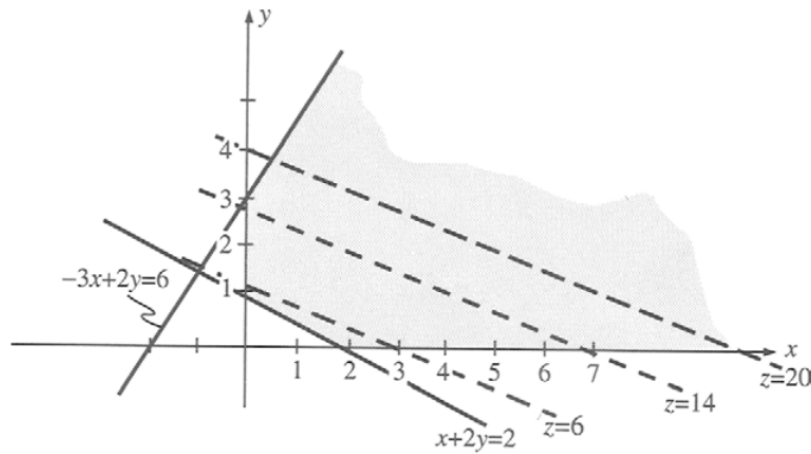


We are able to see that the solution is somewhere between $z = 12$ and $z = 15$. In fact, the solution is when $z = 27/2$, leading to the solution of $(x, y) = (3/2, 5/2)$.



Example 11.2 Consider the linear programming problem

$$\begin{aligned} &\text{Maximize } z = 2x + 5y \text{ subject to} \\ &-3x + 2y < 6 \\ &x + 2y > 2 \\ &x > 0, y > 0 \end{aligned}$$



11.1.1 Feasible Region

We now explore the question of where in the set of feasible solutions we are likely to find a point at which the objective function takes on its optimal value. We first show that if x_1 and x_2 are two feasible solutions, then any point on the line segment joining these two points is also a feasible solution. The line segment joining x_1 and x_2 is defined as

$$\{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = p\mathbf{x}_1 + (1-p)\mathbf{x}_2\}, \quad 0 \leq p \leq 1$$

We define the end points as $\mathbf{x}_1, \mathbf{x}_2$, and any points in between them are interior points

If both of these points are feasible, then they both satisfy the constraints

$$\mathbf{a}^\top \mathbf{x}_1 < b_i, \quad \text{and} \quad \mathbf{a}^\top \mathbf{x}_2 < b_i,$$

so any point in between also satisfy this

$$\begin{aligned}\mathbf{a}^\top \mathbf{x} &= \mathbf{a}^\top (p\mathbf{x}_1 + (1-p)\mathbf{x}_2) \\ &= p\mathbf{a}^\top \mathbf{x}_1 + (1-p)\mathbf{a}^\top \mathbf{x}_2 \\ &\leq pb_i + (1-p)b_i = b_i\end{aligned}$$

Now consider the objective function for two feasible points where both $\mathbf{c}^\top \mathbf{x}_1 = \mathbf{c}^\top \mathbf{x}_2 = k$?

Now lets $\mathbf{c}^\top \mathbf{x}_1 \leq \mathbf{c}^\top \mathbf{x}_2$, then we have

$$\begin{aligned}\mathbf{c}^\top \mathbf{x} &= \mathbf{c}^\top (p\mathbf{x}_1 + (1-p)\mathbf{x}_2) \\ &= p\mathbf{c}^\top \mathbf{x}_1 + (1-p)\mathbf{c}^\top \mathbf{x}_2 \\ &< p\mathbf{c}^\top \mathbf{x}_2 + (1-p)\mathbf{c}^\top \mathbf{x}_2 \\ &= \mathbf{c}^\top \mathbf{x}_2\end{aligned}$$

Definition 11.3 A set C is convex if for all $x, y \in C$ and $p \in [0, 1]$,

$$px + (1-p)y \in C.$$

Proposition 11.4 The feasible region of a linear program is convex.

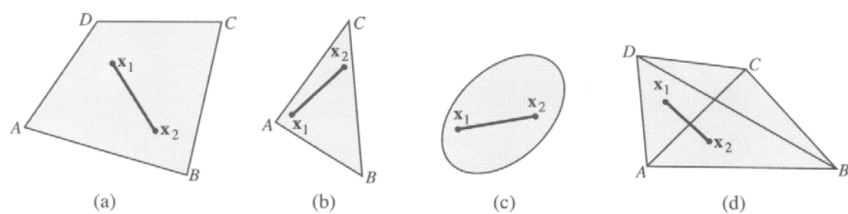


FIGURE 1.9

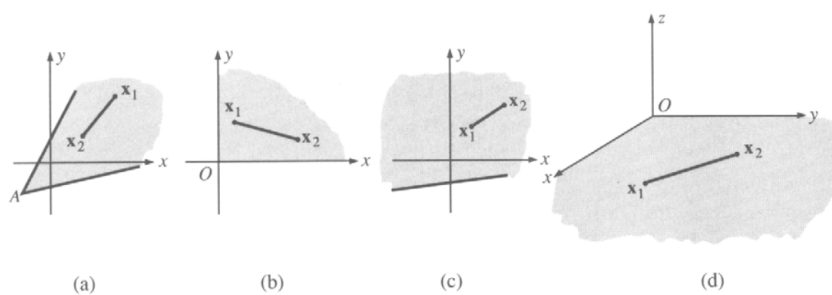


FIGURE 1.10

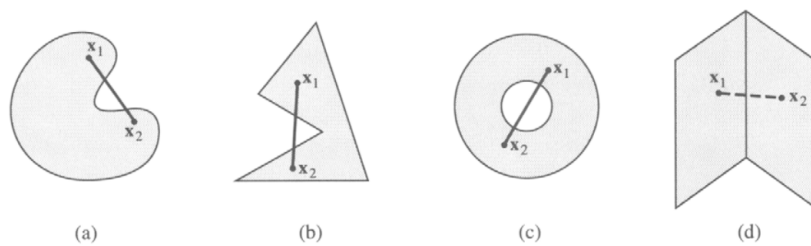


Figure 11.1: